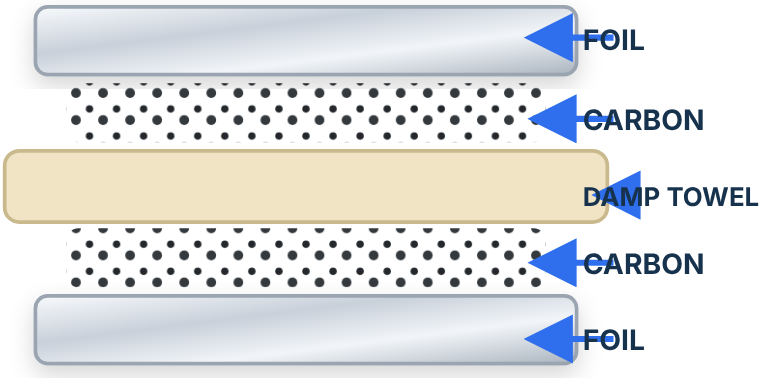
EASY ILLUSTRATED LAB GUIDE

Build a Lemon-Juice Supercapacitor

One picture and one short action per step

ONE CAPACITOR = ONE SANDWICH



Build 4 complete sandwiches and stack them.

Swipe to Step 1

For instructor-supervised high-school use

1

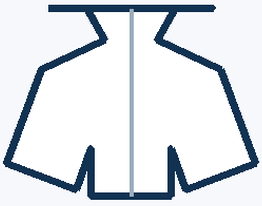
STUDENT PHONE GUIDE

Put on safety gear

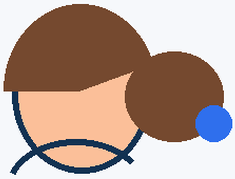
BE READY BEFORE YOU START



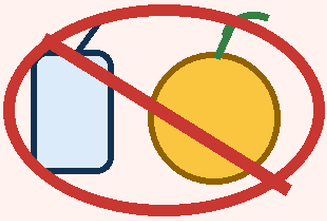
SPLASH GOGGLES



LAB COAT / APRON



TIE BACK LONG HAIR



NO FOOD OR DRINK

Wear goggles and an apron. Tie back long hair. Keep food and drinks away.

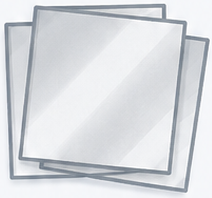
Handle carbon gently - do not blow or scatter it. Connect the 9 V battery only when told.

2

STUDENT PHONE GUIDE

Gather your materials

GET EVERYTHING READY



8 FOIL SQUARES



4 TOWEL SQUARES



LEMON JUICE



ACTIVATED CARBON



RED LED



9 V BATTERY



2 ALLIGATOR LEADS



2 JUMPER WIRES



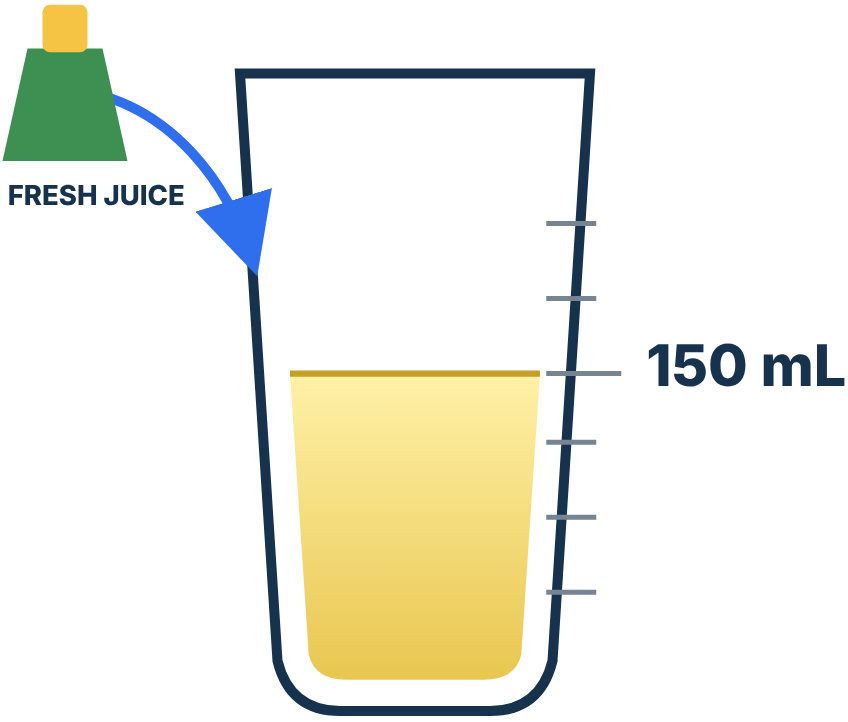
MULTIMETER

Also bring: 250 mL beaker, weight, tray, and carbon scoop.

3

STUDENT PHONE GUIDE

Measure the lemon juice



Pour **150 mL of fresh lemon juice** into the 250 mL beaker.

Keep the juice beaker away from the multimeter and battery.

4

STUDENT PHONE GUIDE

Wet all 4 towel squares

REPEAT THIS FOR ALL 4 TOWEL SQUARES

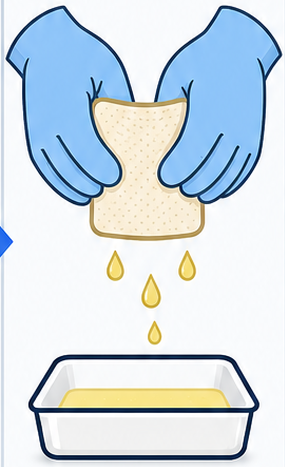
1



SOAK

Wet the whole square

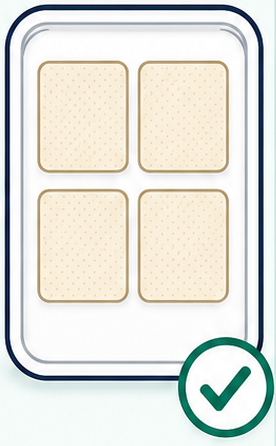
2



SQUEEZE

Remove extra juice

3



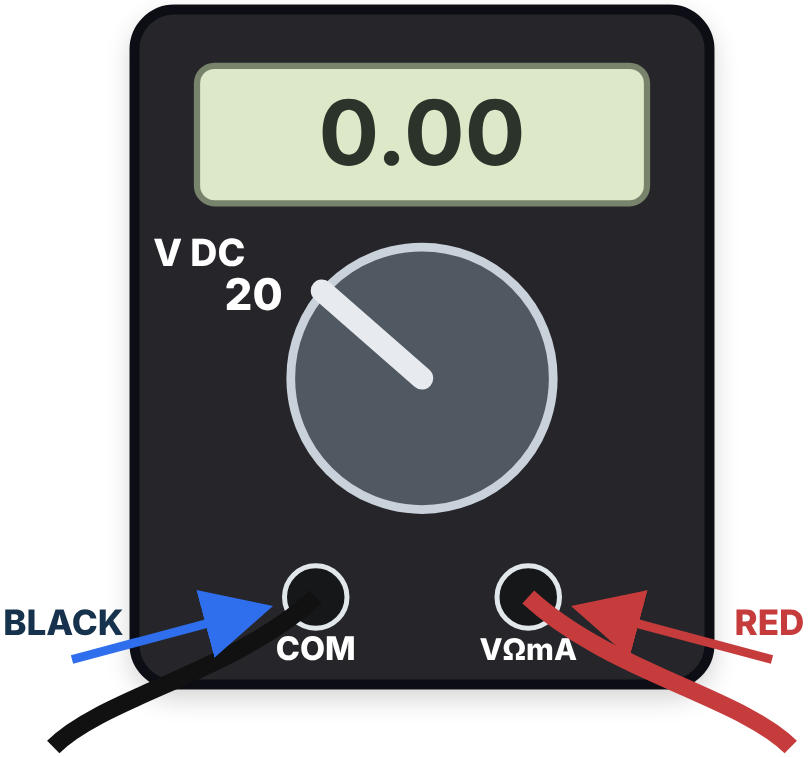
SET IN TRAY

Damp - not dripping

5

STUDENT PHONE GUIDE

Set the multimeter now



Black probe goes in **COM**.

Red probe goes in **VΩmA** or the voltage port.

Turn the dial to **20 V DC** (V with a straight/dashed line).

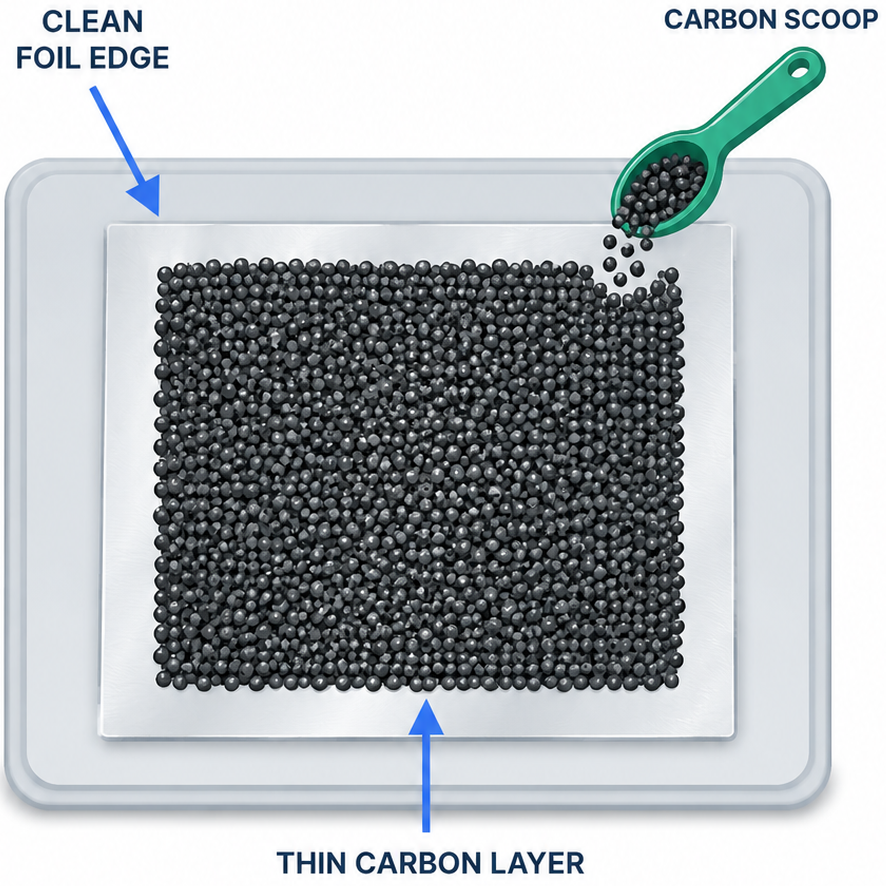
Leave the probes ready. The stored voltage falls very quickly.

6

STUDENT PHONE GUIDE

Start capacitor #1

LEAVE A CLEAN BORDER

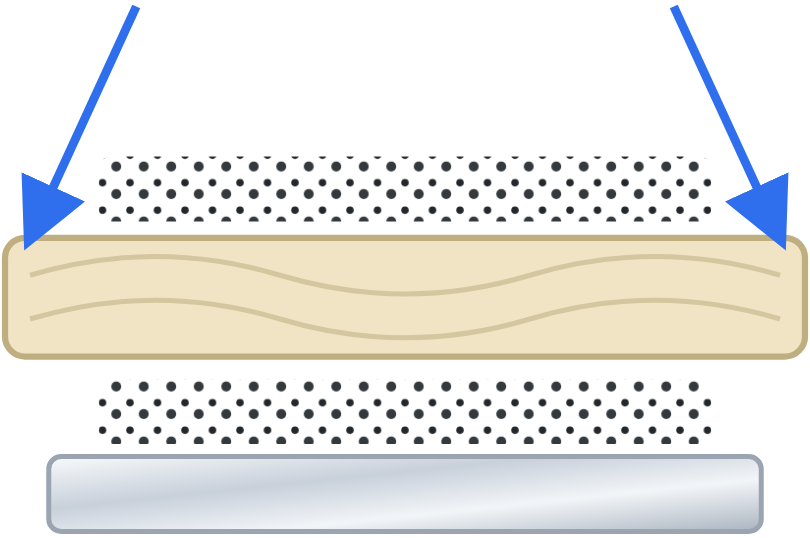


! Do not let carbon reach the foil edge.

STUDENT PHONE GUIDE

7 Add towel and more carbon

TOWEL MUST STICK OUT PAST THE FOIL



FOIL - CARBON - DAMP TOWEL - CARBON

Center one damp towel square over the carbon.

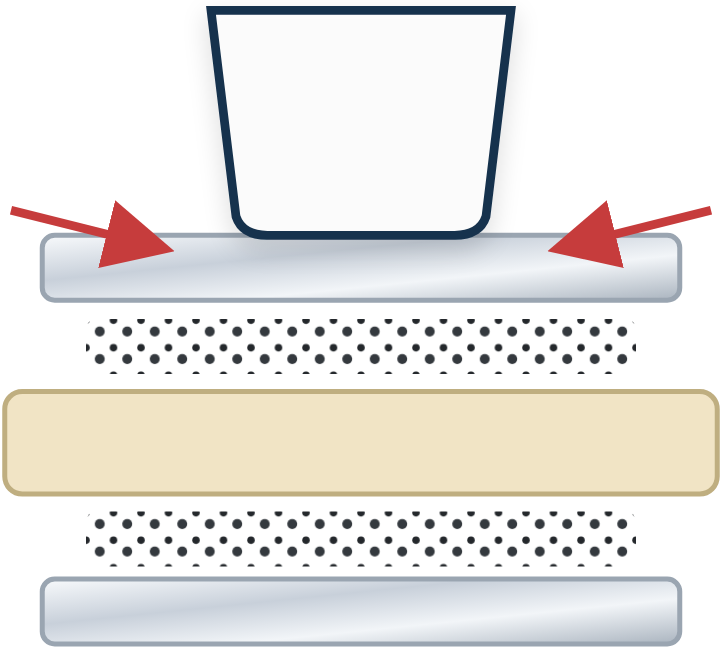
The towel should stick out past the foil on every side.

Add another thin, even carbon layer on top of the towel.

8

STUDENT PHONE GUIDE

Finish the first sandwich



CHECK: THE TWO FOILS DO NOT TOUCH

Place a second foil square on top.

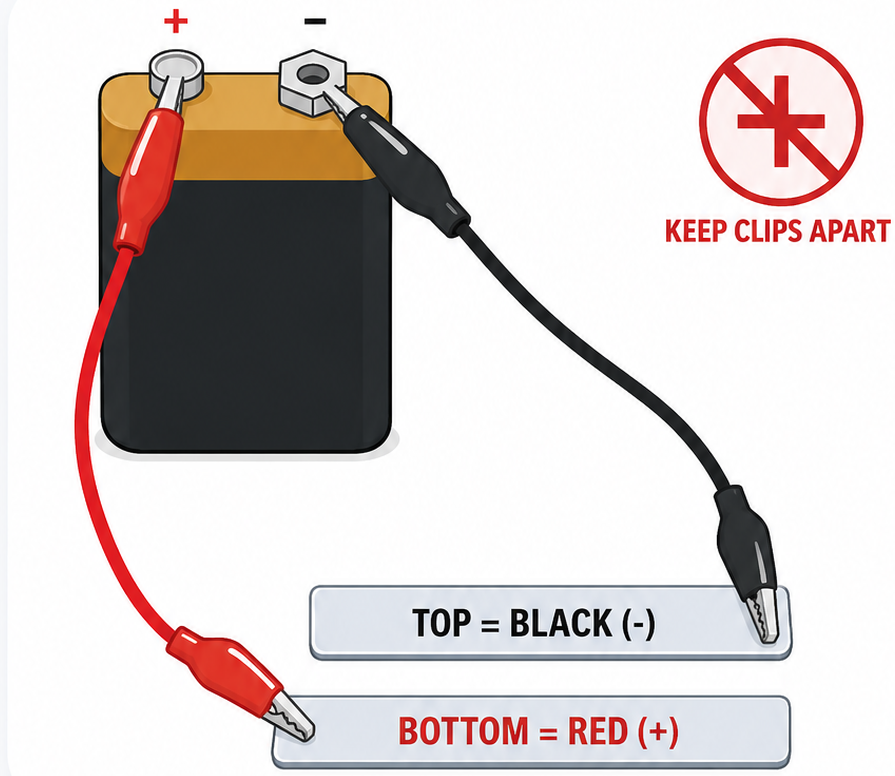
Check that the top and bottom foil **do not touch** around the edges.

Set the beaker or clean weight on top so the layers stay pressed together.

One complete capacitor is: foil - carbon - damp towel - carbon - foil.

STUDENT PHONE GUIDE

9 Connect the 9 V battery



Red clip: small round **positive** (+) battery terminal to the **bottom** foil.

Black clip: large hexagonal **negative** (-) terminal to the **top** foil.

Make sure the two metal clips never touch each other.

10

STUDENT PHONE GUIDE

Charge for 2 minutes

WATCH THE SETUP THE WHOLE TIME



- Start a 2-minute timer.
- Watch the battery, wires, and capacitor the whole time.
- Do not move the stack while it charges.

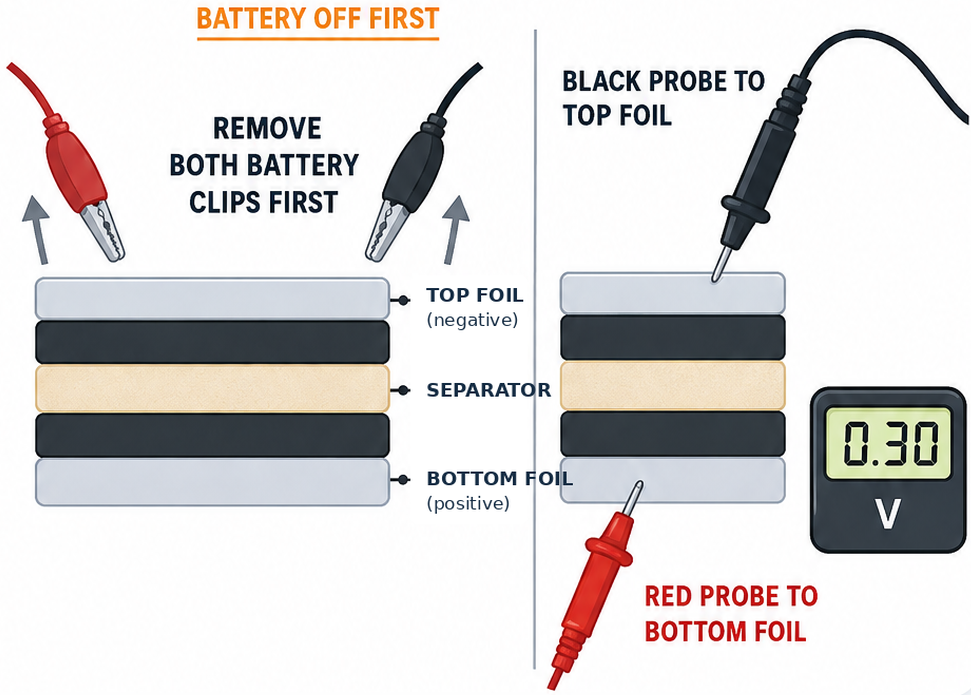
Stop and tell the instructor if anything becomes warm, fizzes, smells unusual, sparks, or leaks badly.

11

STUDENT PHONE GUIDE

Disconnect and measure FAST

WRITE THE FIRST NUMBER YOU SEE



1. At 2 minutes, remove both battery clips from the foil.
2. Immediately touch the **red** meter probe to the **bottom foil.**
3. Touch the **black** meter probe to the **top foil.**
4. **Write the first number you see.**

The number may fall toward 0 almost immediately. That is expected.

12

STUDENT PHONE GUIDE

Record voltage #1

CAPACITORS	VOLTAGE
1 →	____ V
2	____ V
3	____ V
4	____ V

A small number can still mean it worked.

Write the first reading in the row for 1 capacitor.

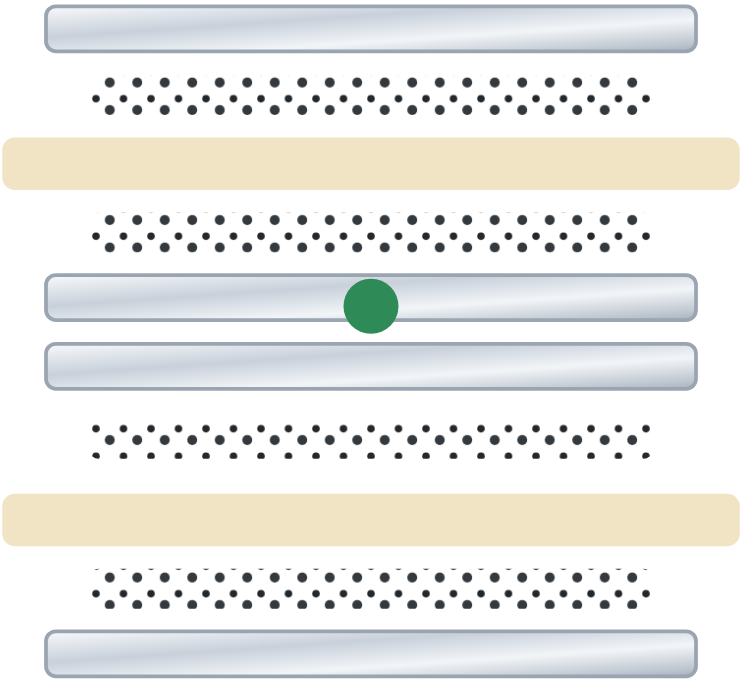
A small number, such as a few tenths of a volt, can still mean the capacitor stored charge.

13

STUDENT PHONE GUIDE

Add capacitor #2

THE NEW BOTTOM FOIL TOUCHES THE OLD TOP FOIL



USE 2 NEW FOIL PIECES FOR CAPACITOR #2

Leave capacitor #1 assembled.

Place a **fresh foil square** directly on the old top foil.

Then add: carbon - damp towel - carbon - another fresh foil.

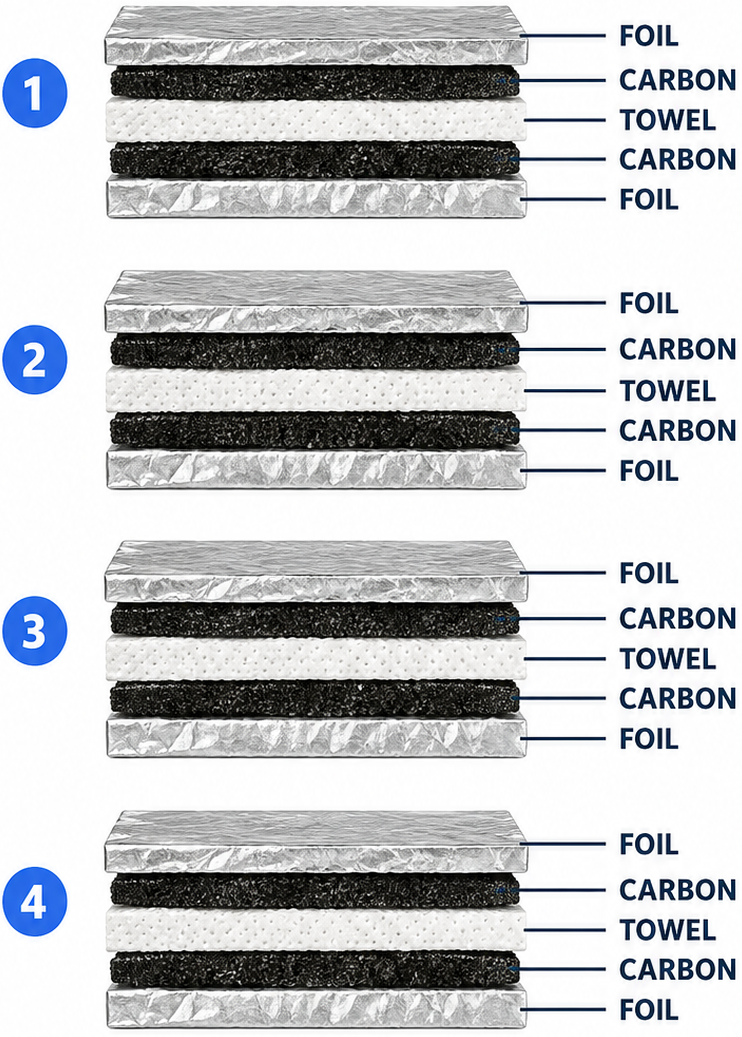
Replace the weight, charge 2 minutes, measure, and record.

Do not reuse the old top foil as the new bottom foil. Each capacitor uses 2 fresh foil pieces.

14

STUDENT PHONE GUIDE

Repeat until you have 4



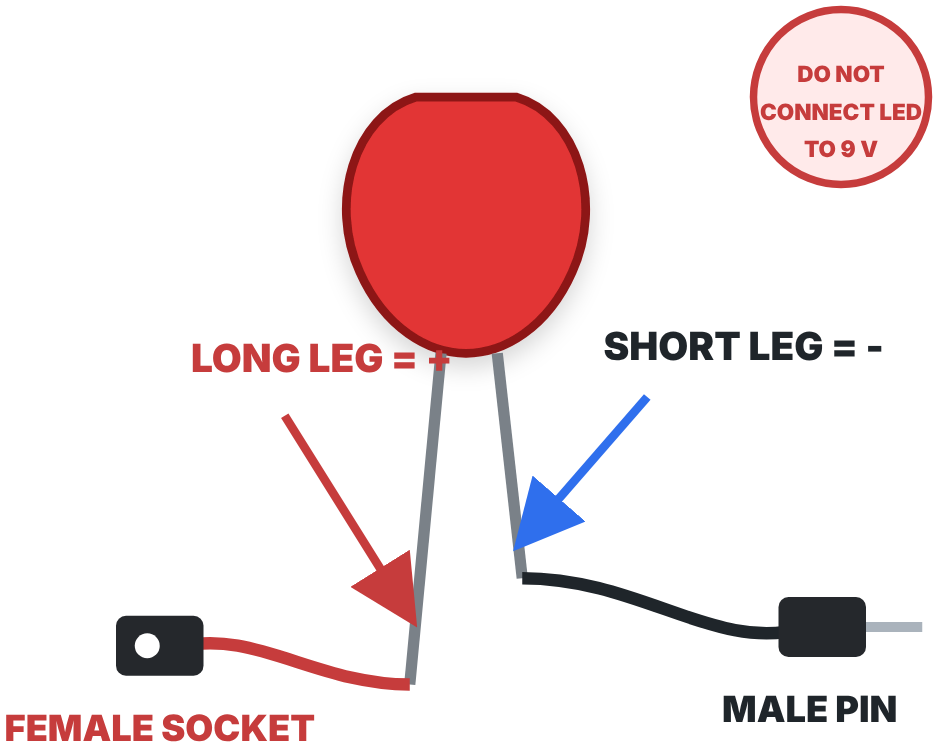
AFTER EACH NEW CAPACITOR:

WEIGHT -> CHARGE 2 MIN -> MEASURE -> RECORD

15

STUDENT PHONE GUIDE

Prepare the red LED



Push one **female socket** onto each LED leg.

The **long LED leg is positive (+)**. The short leg is negative (-).

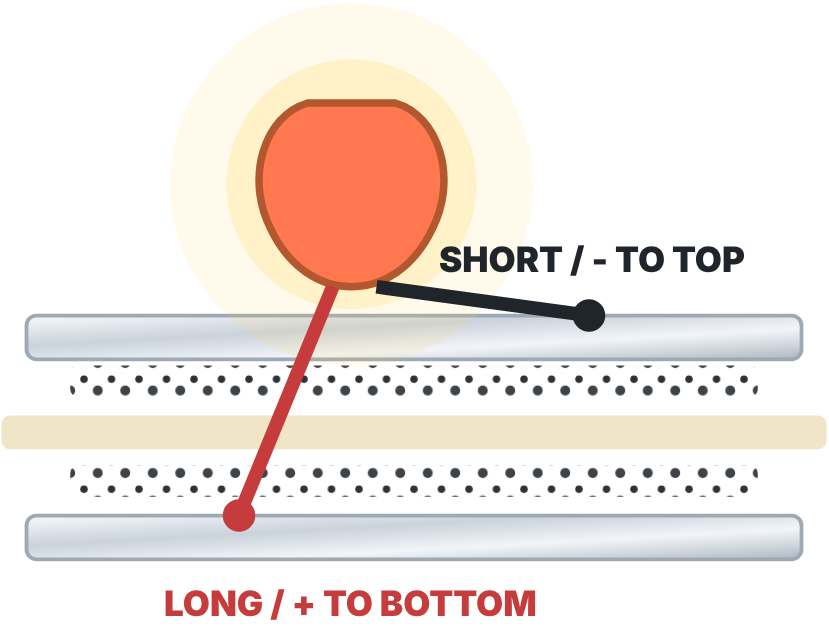
The exposed male pins will touch the foil.

Never connect the bare LED directly to the 9 V battery.

16

STUDENT PHONE GUIDE

Charge and test the LED



WATCH CLOSELY - THE FLASH MAY BE VERY BRIEF

Charge the completed 4-capacitor stack for 2 minutes.

Disconnect the battery and quickly measure the voltage.

Touch the wire from the **long LED leg to the bottom foil**.

Touch the wire from the **short leg to the top foil**.

Look closely. The LED may only flash for a moment. Only try a 3-minute recharge if the instructor approves.

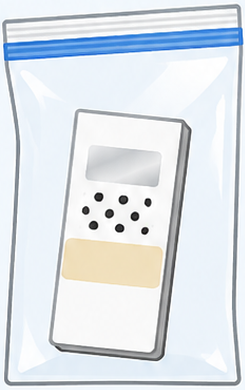
17

STUDENT PHONE GUIDE

Record and clean up

CLEAN UP IN THIS ORDER

1



BAG THE USED STACK

2



DAMP-WIPE THE TRAY

3



WASH YOUR HANDS

DO NOT SWEEP, BLOW, OR RINSE LOOSE CARBON